Math 1232 Summer 2025
Mastery Quiz 11
Due Wednesday, August 6

This mastery quiz has three topics. Everyone should do S8, but if you already have a 4/4 on M3 or M4, feel free to skip these topics. Please do your best, but don't worry if you make a minor error—your goal is to demonstrate your mastery of the underlying material.

Feel free to consult your notes, but please don't discuss the actual quiz questions with other students in the course.

Remember that you are trying to demonstrate that you understand the concepts involved. For all these problems, justify your answers and explain how you reached them. Do not just write "yes" or "no" or give a single number.

Please turn in this quiz in class on Wednesday. You may print this document out and write on it, or you may submit your work on separate paper; in either case make sure your name is clearly on it. If you absolutely cannot turn it in in person, you can submit it electronically but this should be a last resort.

Topics on this quiz:

- Major Topic 3: Series Convergence
- Major Topic 4: Taylor Series
- Secondary Topic 8: Parameterization

Name:

Name: ______________________________

Major Topic 3: Series Convergence

Remember to show *all* your work—this includes stating any convergence tests you use, as well as computing any limits that arise.

(a) Analyze the convergence of the series $\displaystyle\sum_{n=1}^{\infty}(-1)^n\left(\frac{2n^2+n+1}{n^2-3n+2}\right)^n$.

(b) Analyze the convergence of the series $\displaystyle\sum_{n=1}^{\infty}\sqrt{n^3+1}$.

Name:

(c) Analyze the convergence of the series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{3n+2}$.

Name:

Major Topic 4: Taylor Series

(a) Using series we already know, write down a formula for the (infinite) Taylor series for $x^2 \ln(1-2x^3)$, and then write down the degree-eleven polynomial explicitly.

(b) Using series we already know, write down a formula for the (infinite) Taylor series for $x(8+x)^{5/3}$, and then write down the degree-four polynomial explicitly.

Name:

(c) Estimate the error if you use $T_4(x) = 1 - \frac{x^2}{2} + \frac{x^4}{4!}$ to approximate $g(x) = \cos(x)$ at $x = -1/2$.

Name: ____

Secondary Topic 8: Parameterization

(a) Find a (implicit, cartesian, not parametric) equation for a line tangent to the polar curve given by $r = 2 + \cos(\theta)$ at the polar coordinates $(5/2, \pi/3)$.

(b) Find the length of the curve parametrized by $x = 3t^2, y = 3t - t^3$, for $1 \leq t \leq 4$.